

PatientTriage.ai: Business Proposal & Enterprise Strategy

“Closing the Emergency Waiting-Room Surveillance Gap”

Our Idea: PatientTriage.ai helps Emergency Department staff notice which waiting patients may be getting worse and need attention first.

Gross Annual Value \$3.82M / yr	Net Annual ROI \$3.58M / yr (14.9x)	LWBS Recovery +\$1.12M / yr (30% drop)	Target Market 5,500+ US/EU EDs
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1. The Business Problem

Emergency Departments are extremely busy. Triage helps doctors and nurses understand a patient's condition when they arrive, but triage is only a snapshot. A patient may then wait for 2.5–4.5 hours unmonitored. During that period, condition can change, vital signs can become abnormal, data can become old, or a monitor can disconnect. Staff may not immediately notice the change, creating a dangerous surveillance gap between triage and treatment. **PatientTriage.ai is designed to help hospitals manage this gap.**

2. Our Solution: Extra Safety Layer for the Waiting Room

PatientTriage.ai continuously looks at available patient information and answers three simple questions:

- **1. Is the patient getting worse?** Checks changes in vital signs over time (ΔSpO_2 , ΔHR).
 - **2. Is the information still reliable?** Checks how old the latest observations are (shelf-life expiration).
 - **3. Do we know enough about the patient?** If important information is missing, it asks staff to physically verify the patient (*Unknown* $\neq$ *Safe*).
- Then it creates a simple focused list: **Who needs attention first?**

3. Simple User Flow & Decision Architecture

Patient Arrives ■■■ Intake Triage ■■■ Waiting Lounge ■■■ Continuous Check ■■■ 4 States (■/■/■/■) ■■■ Priority Queue ■■■ Clinician Decides

The system supports the clinician. It does not replace the clinician. Medical professionals retain 100% decision authority.

4. Value for Hospital Staff & Leadership

Role	Current Problem	PatientTriage.ai Value Delivery
Triage Nurses (RNs)	Overwhelmed tracking 40+ waiting patients; alarm fatigue.	Focused 'Next 5 Minutes' queue highlights patients needing immediate attention.
Emergency Doctors	Blind to which waiting patients deteriorated since intake.	Instant visibility into who has deteriorated, needs reassessment, or has warning signs.
Hospital Leadership	Delayed diagnosis risks, sentinel waiting lounge events.	Better waiting-room visibility, fewer missed deteriorations, better clinical audit trail.
Finance Teams	LWBS walkout revenue losses, uncompensated ICU transfers.	Captures modeled \$3.58M net annual value from recovered walkouts & fewer ICU crashes.

5. Business Impact & Value Breakdown (Modeled 500-Bed Facility)

Model based on 65,000 annual ED visits, 500 beds, and \$1,200 average baseline ED revenue per visit (projections from business model, not actual hospital results):

Area of Value	Expected Operational Impact	Modeled Annual Value
1. Fewer patients leaving without being seen (LWBS)	30% reduction via proactive re-engagement	\$1,123,200 / yr
2. Fewer waiting-room ICU escalations	93 avoided stays (@ \$15k per stay)	\$1,395,000 / yr
3. Lower malpractice liability risk	40% reduction via deterministic audit logs	\$480,000 / yr
4. Staff retention & overtime savings	Lower nurse turnover + 15% overtime reduction	\$378,000 / yr
5. Better ED throughput & boarding	30-minute improvement via optimized dispatch	\$445,000 / yr
Total Gross Annual Value	—	\$3,821,200 / yr
Modeled Software & Hardware Subscription	Enterprise annual software license + support	-\$240,000 / yr
Net Estimated Annual Value (14.9x ROI)	Net Return on Investment	+\$3,581,200 / yr

6. Hospital Integration, Safety Rules & Fail-Safe Design

- **HL7 FHIR Integration:** Works alongside existing Electronic Health Record (EHR) systems using standard data such as Patient Encounters and Vital Observations. Local edge processing avoids cloud AI latency.
- **Safety & Fixed Thresholds:** Serious conditions (SpO₂ < 85%, SBP < 75 mmHg) directly trigger an escalation without being masked by a statistical score.
- **Fail-Safe Behavior:** If network connectivity or telemetry fails, the system prompts staff to perform manual checks while local safety alerts remain active.

7. Target Market & Implementation Roadmap

- **Target Market:** 5,500+ Emergency Departments across US & Europe with high patient volumes and long wait times.
- **Stage 1 — Prototype (Completed - Q3 2026):** Built and tested with 51 automated tests passing.
- **Stage 2 — Shadow Deployment (Q4 2026):** Run alongside existing hospital processes to compare recommendations with clinician decisions.
- **Stage 3 — Pilot (Q1–Q2 2027):** Deploy in ED setting to measure time to escalation, wait times, LWBS, and alert usefulness.
- **Stage 4 — Scale (Q3 2027+):** Expand to more hospitals, ambulance telemetry, and regional patient flow coordination.

Final Proposition: PatientTriage.ai is not another triage system. It is a waiting-room monitoring and prioritization layer that works after triage and before treatment, answering: “Who has changed, and who needs attention now?”

Because a patient being stable at 2:00 PM does not mean they will still be stable at 4:00 PM. Triage is a snapshot. Risk isn't.